ENGINEERING DECISION SUPPORT

SYNTHETIC UNIT

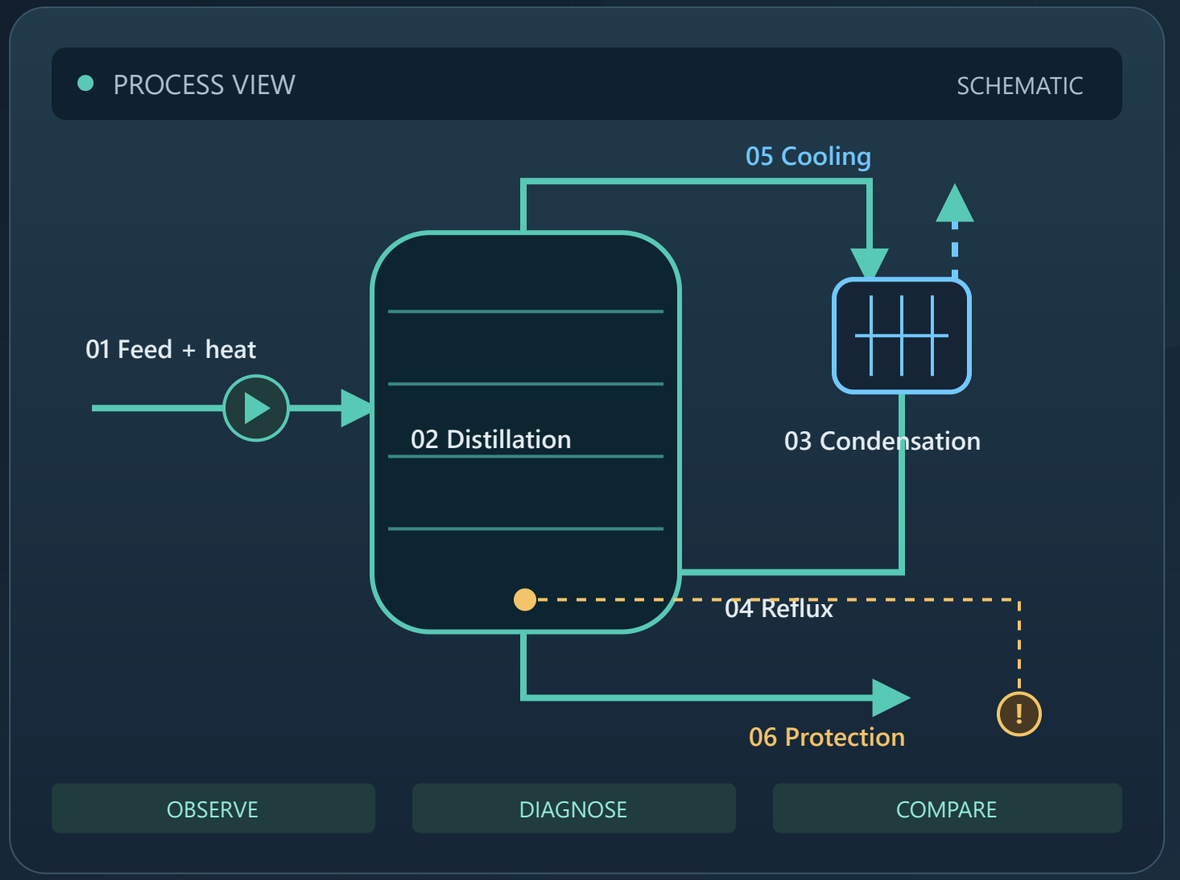
# REFINERY

See the signals.

**Compare the consequences.**

Six loops. Observed evidence.
Explainable hypotheses. What-if forecasts.

**CPU DEMO** HUMAN REVIEW




Synthetic process simulation · No equipment actuation

AMD Token Factory integration pending

# Signals need a decision context.

A temperature rise is an observation. Deciding what to inspect next needs a connected view of the process.

01

### See the unit

Follow temperatures, pressures, alarms, and nominal references across six logical loops.

02

### Inspect the evidence

Review competing diagnostic hypotheses and the observations behind them.

03

### Compare consequences

Explore conditional action forecasts before a human review.

Demonstration scope: a synthetic refinery unit, using simulated observations.

# From observations to review.

One calculation connects a process scenario to an inspectable engineering workflow.

**SIMULATE**

### Physical model

Nominal and disturbed trajectories

→

**OBSERVE**

### Available signals

Temperatures, pressures, protection

→

**DIAGNOSE**

### Rank hypotheses

Evidence and similarity scores

→

**COMPARE**

### What-if forecasts

Assumptions and action consequences

**Reviewable output**

Persisted calculation history, charts, notes, and exports.

**Clear information boundary**

Hidden simulator truth is separate from the diagnostic observation view.

# Six loops. One shared view.

- 01 Feed + heating
- 02 Distillation
- 03 Condensation
- 04 Reflux return
- 05 Cooling
- 06 Independent protection

## Built for inspection

Expand all equipment signals. Align charts in time. Review alarms and engineering notes.

## Built for presentation

Dedicated video wall, loop summaries, histograms, and shared-time trends.

EN / RU · Light / dark · Configurable labels

> Loop summaries describe available observations; they do not invent unmeasured utility data.

# Investigate a cooling disturbance.

The user selects a synthetic scenario. Diagnosis then works from the resulting observations.

1

### Run + inspect

Start a cooling-loss scenario. Follow pressure and temperature against the nominal reference.

2

### Review hypotheses

Open ranked hypotheses, compare their evidence, and inspect the selected analysis time.

3

### Compare actions

Compare hold, restore cooling, reduce heating, and reduce feed under stated assumptions.

Conditional model forecasts support review. The prototype issues no equipment commands.

# Make the limits visible.

## What the prototype demonstrates

- A working synthetic process simulation and operator interface.
- Observation-based hypothesis ranking and action forecasts.
- Persisted results, engineering review, and report exports.
- A limited A/B/C synthetic model experiment with integrity checks.

## What still needs validation

- Similarity scores are not calibrated probabilities.
- Forecasts depend on model assumptions.
- Imported data has unverified provenance.
- Real-plant calibration, authenticated integration, and industrial validation remain future work.

> Missing observations remain missing. Display alarm settings do not change physical protection.

# A reproducible CPU demo today.

## Implemented stack

Python

NumPy · SciPy · CoolProp

JavaScript · HTML · CSS

Local server and browser workspace.
No API key is required for the CPU demo.

PENDING

## AMD Token Factory integration

Account registration and an official endpoint/model are still required.

**No AMD inference or GPU execution is claimed.**

Proposed track: Intelligent Industry

# Run. Inspect. Compare.

REFINERY turns a synthetic disturbance into an evidence-led engineering review.

```
python -m amd.server
```

Open http://127.0.0.1:8765/industry

### 1 / Run

Create a new calculation with action comparison.

### 2 / Inspect

Follow signals, evidence, and ranked hypotheses.

### 3 / Compare

Review conditional forecasts and the video wall.

github.com/CAPEXNULL/REFINERY-ACT-III · capexnull.github.io/REFINERY-ACT-III/
Permanent package: recorded replay /demo/ · narrated video /demo.mp4 (2:28)
Temporary live CPU demo (host must be running). AMD Token Factory integration pending; no AMD inference claimed.